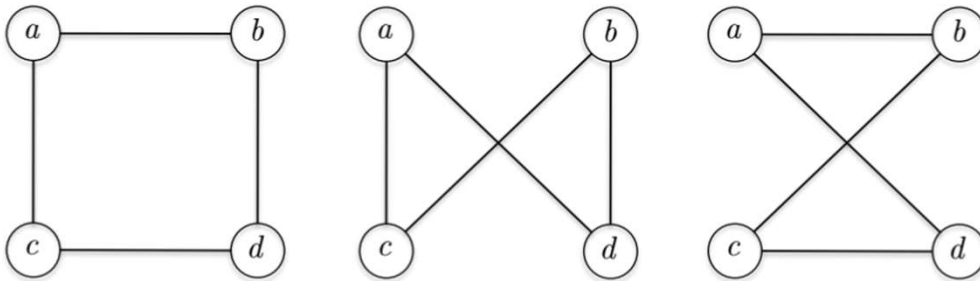


Travelling Salesman Problem

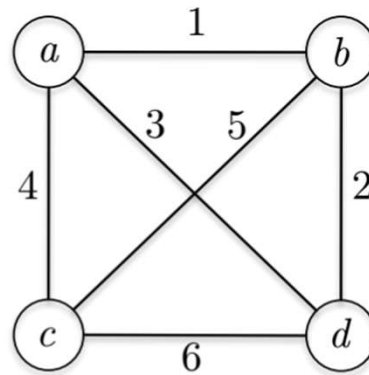
Tours among cities a,b,c,d

- a,b,d,c equivalent to b,d,c,a (change in start vertex)
- a,b,c,d $\Rightarrow$ d,c,b,a (reversing the tour)



Quiz 19.2

What is the minimum sum of edge costs of a tour of the following graph? (Each edge is labeled with its cost.)



- a) 12
- b) 13
- c) 14
- d) 15

Correct answer: (b). We can enumerate tours by starting with the vertex a and trying all six possible orderings of the other three vertices, with the understanding that the tour finishes by traveling from the last vertex back to a . (Actually, this enumeration counts each tour twice, once in each direction.) The results:

Vertex Ordering	Cost of Corresponding Tour
a, b, c, d or a, d, c, b	15
a, b, d, c or a, c, d, b	13
a, c, b, d or a, d, b, c	14

The shortest tour is the second one, with a total cost of 13.

Traveling Salesman Problem

- Minimum Spanning Tree

$G(V, E) \Rightarrow$ Spanning Tree by removing edges to get a tree with V .

$\Rightarrow$ Minimum cost of edges among all spanning trees

Kruskal, Prim $\Rightarrow O((m+n) \log n)$ with m edges and n vertices.

The traveling-salesman problem

In the traveling-salesman problem introduced in Section 34.5.4, we are given a complete undirected graph $G = (V, E)$ that has a nonnegative integer cost $c(u, v)$ associated with each edge $(u, v) \in E$, and we must find a hamiltonian cycle (a tour) of G with minimum cost. As an extension of our notation, let $c(A)$ denote the total cost of the edges in the subset $A \subseteq E$:

$$c(A) = \sum_{(u,v) \in A} c(u, v) .$$

In many practical situations, the least costly way to go from a place u to a place w is to go directly, with no intermediate steps. Put another way, cutting out an intermediate stop never increases the cost. We formalize this notion by saying that the cost function c satisfies the ***triangle inequality*** if, for all vertices $u, v, w \in V$,

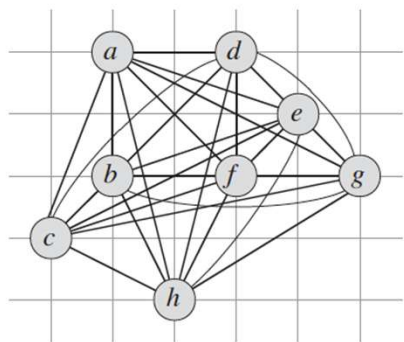
$$c(u, w) \leq c(u, v) + c(v, w) .$$

The triangle inequality seems as though it should naturally hold, and it is automatically satisfied in several applications. For example, if the vertices of the graph are points in the plane and the cost of traveling between two vertices is the ordinary euclidean distance between them, then the triangle inequality is satisfied.

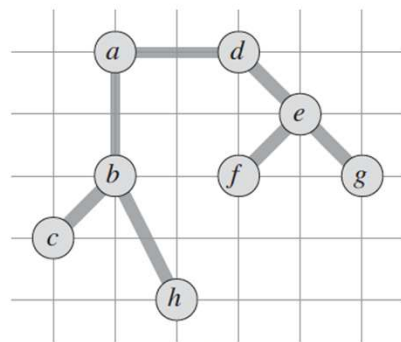
With triangle inequality

APPROX-TSP-TOUR(G, c)

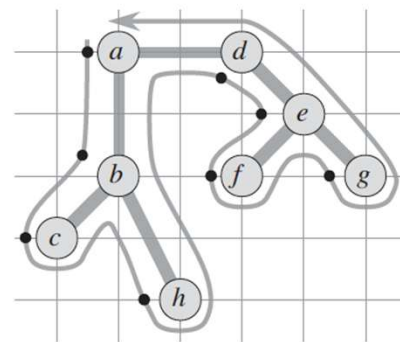
- 1 select a vertex $r \in G.V$ to be a “root” vertex
- 2 compute a minimum spanning tree T for G from root r
using MST-PRIM(G, c, r)
- 3 let H be a list of vertices, ordered according to when they are first visited
in a preorder tree walk of T
- 4 **return** the hamiltonian cycle H



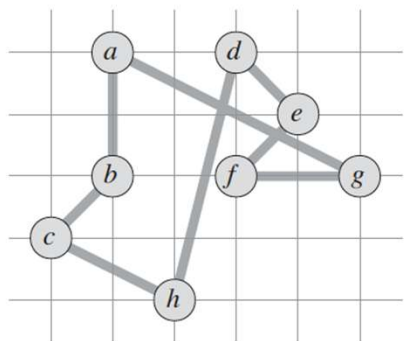
(a)



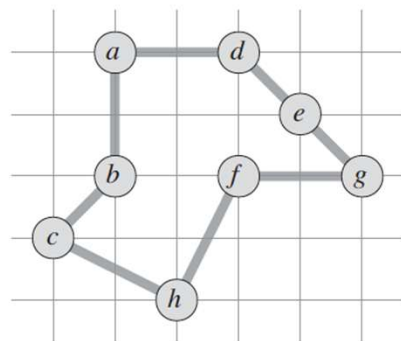
(b)



(c)



(d)



(e)

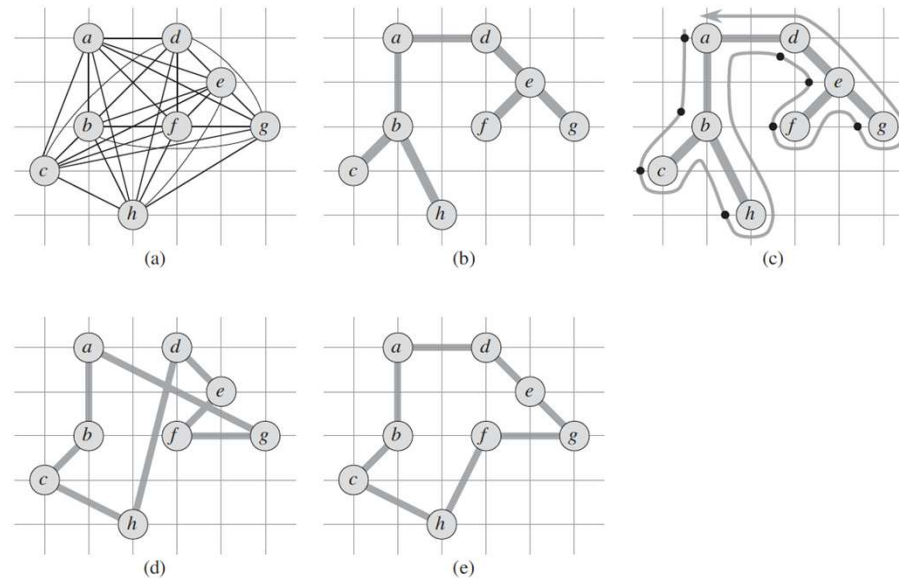


Figure 35.2 The operation of APPROX-TSP-TOUR. **(a)** A complete undirected graph. Vertices lie on intersections of integer grid lines. For example, f is one unit to the right and two units up from h . The cost function between two points is the ordinary euclidean distance. **(b)** A minimum spanning tree T of the complete graph, as computed by MST-PRIM. Vertex a is the root vertex. Only edges in the minimum spanning tree are shown. The vertices happen to be labeled in such a way that they are added to the main tree by MST-PRIM in alphabetical order. **(c)** A walk of T , starting at a . A full walk of the tree visits the vertices in the order $a, b, c, b, h, b, a, d, e, f, e, g, e, d, a$. A preorder walk of T lists a vertex just when it is first encountered, as indicated by the dot next to each vertex, yielding the ordering a, b, c, h, d, e, f, g . **(d)** A tour obtained by visiting the vertices in the order given by the preorder walk, which is the tour H returned by APPROX-TSP-TOUR. Its total cost is approximately 19.074. **(e)** An optimal tour H^* for the original complete graph. Its total cost is approximately 14.715.

Theorem 35.2

APPROX-TSP-TOUR is a polynomial-time 2-approximation algorithm for the traveling-salesman problem with the triangle inequality.

Proof We have already seen that APPROX-TSP-TOUR runs in polynomial time.

Let H^* denote an optimal tour for the given set of vertices. We obtain a spanning tree by deleting any edge from a tour, and each edge cost is nonnegative. Therefore, the weight of the minimum spanning tree T computed in line 2 of APPROX-TSP-TOUR provides a lower bound on the cost of an optimal tour:

$$c(T) \leq c(H^*) . \quad (35.4)$$

A **full walk** of T lists the vertices when they are first visited and also whenever they are returned to after a visit to a subtree. Let us call this full walk W . The full walk of our example gives the order

$a, b, c, b, h, b, a, d, e, f, e, g, e, d, a$.

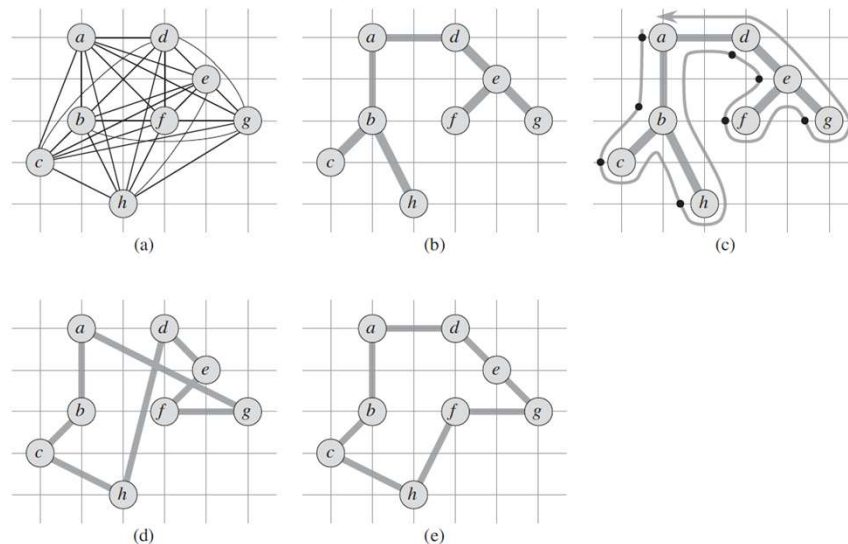
Since the full walk traverses every edge of T exactly twice, we have (extending our definition of the cost c in the natural manner to handle multisets of edges)

$$c(W) = 2c(T) . \quad (35.5)$$

Inequality (35.4) and equation (35.5) imply that

$$c(W) \leq 2c(H^*) , \quad (35.6)$$

and so the cost of W is within a factor of 2 of the cost of an optimal tour.



Unfortunately, the full walk W is generally not a tour, since it visits some vertices more than once. By the triangle inequality, however, we can delete a visit to any vertex from W and the cost does not increase. (If we delete a vertex v from W between visits to u and w , the resulting ordering specifies going directly from u to w .) By repeatedly applying this operation, we can remove from W all but the first visit to each vertex. In our example, this leaves the ordering a, b, c, h, d, e, f, g .

This ordering is the same as that obtained by a preorder walk of the tree T . Let H be the cycle corresponding to this preorder walk. It is a hamiltonian cycle, since ev-

Since the full walk traverses every edge of T exactly twice, we have (extending our definition of the cost c in the natural manner to handle multisets of edges)

$$c(W) = 2c(T) . \tag{35.5}$$

Inequality (35.4) and equation (35.5) imply that

$$c(W) \leq 2c(H^*) , \tag{35.6}$$

and so the cost of W is within a factor of 2 of the cost of an optimal tour.

ery vertex is visited exactly once, and in fact it is the cycle computed by APPROX-TSP-TOUR. Since H is obtained by deleting vertices from the full walk W , we have

$$c(H) \leq c(W) . \tag{35.7}$$

Combining inequalities (35.6) and (35.7) gives $c(H) \leq 2c(H^*)$, which completes the proof. ■